

Fabricating Haptic Paper Interface

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ABSTRACT

Our research focuses on developing haptic feedback mechanisms for paper interfaces. We present a comprehensive overview of our efforts and introduce approaches to incorporating haptic feedback into paper interfaces. Traditional paper interfaces primarily emphasize interactive features and often disregard haptic feedback, which is critical for enhancing user confidence in operation. Our research investigates how cardboard structures and static magnets can be utilized to incorporate haptic feedback features into paper interfaces. These techniques help to enhance the user experience associated with paper interfaces.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods.**

KEYWORDS

Paper interface; haptic; sensation feedback; fabrication

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1 INTRODUCTION

Paper interfaces are easy to build and are favored by designers and researchers. However, current interfaces often lack the ability to provide dynamic and interactive sensation feedback. Such sensation feedback is critical for input interfaces, it provides confirmation to the user that their input has been recognized. At the same time, such feedback makes interactions more engaging and enjoyable, creating a more immersive experience for the user. We discuss the techniques we've developed to generate sensation feedback on paper interfaces, including haptic feedback mechanisms based on cardboard structures, as well as magnet-driven sensation feedback mechanisms. By proposing new fabrication methods, we aim to explore ways to enhance the user experience of the paper interface.

2 RELATED WORK

Our work draws inspiration from research on paper-based interactions. Previous studies have used paper as a medium to construct electronic products and prototype interface designs, with paper circuit-based interactions being the most common approach [Kawahara et al. 2013; Qi and Buechley 2010; Zheng et al. 2019]. Techniques to support paper engineering have also been proposed, facilitating the design of paper interfaces [Annett et al. 2015]. These explorations highlight the diverse range of methods and techniques available for building paper interfaces. While there is less work discussing haptic feedback for paper interfaces, Chang's work shows us the importance of such techniques and proposes a haptic feedback mechanism based on the structure of paper [Chang et al. 2020]. Building on these prior works, we explore the creation of haptic feedback with paper. We aim to extend the interactive capabilities of paper as a material and enhance the user experience of paper interfaces.

3 CREATE HAPTIC FEEDBACK WITH CARDBOARD STRUCTURES

We noticed the impact of the lack of haptic feedback on the user experience while developing the Home-based DIY Interactive Information Physicalization project [Bae et al. 2021, 2022]. This project was focused on helping children learn about data physicalization [Bae et al. 2021]. Initially, children used the Tearing Chart (Figure 1) to input data by tearing paper strips, which provided a tactile sensation but was irreversible. The next design, Touching Trends (Figure 2), involved sliding paper modules but often led to inaccurate inputs due to the lack of force feedback.

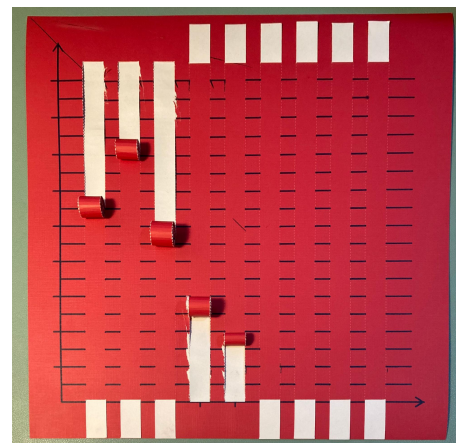


Figure 1: Tearing Chart: tearing off strips of paper



Figure 2: Touching Trends: sliding the paper modules



Figure 3: Data-is-Yours kit: a tangible interface based on cardboard and 3D printed parts. We created haptic feedback on it by changing the cutting direction of the cardboard.

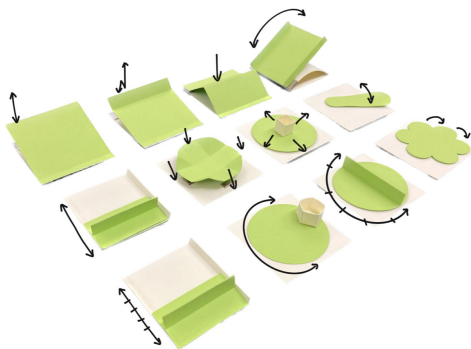


Figure 4: Paper Fidget: create sensation feedback on paper interfaces by attaching neodymium magnets to papers

To address the lack of haptic feedback, we then designed the Data-is-Yours kit [Bae et al. 2022], using cardboard with directional cuts and 3D-printed knobs (Figure 3). By adjusting the orientation of these cuts, we could control the alignment of the cardboard flutes in relation to the sliding direction. When the flutes are in the same direction as the sliding direction, the movement is smooth; when the flutes are perpendicular to the sliding direction, the user will feel the bumps due to friction.

4 CREATE HAPTIC FEEDBACK WITH MAGNETS

In the Paper Fidget project, we attached neodymium magnets on paper, enabling a variety of haptic feedback scenarios without complex paper crafting [Yang 2024]. Figure 4 shows the 12 different units of Paper Fidget. By utilizing magnets as a means to enable mechanical movements on paper, the prototyping and fabrication process of paper interactions becomes simplified. By mounting magnets at different positions on the paper, we can create various haptic feedbacks through the vibrations brought by the magnets, thus enhancing user interaction.

5 DISCUSSION AND CONCLUSION

Paper interfaces are easy to build and give designers and practitioners a platform for rapid fabrication. By adding haptic feedback, our design and fabrication techniques help enhance the user experience of paper interfaces. Our techniques are easy to learn, do not require specialized skills and equipment, and can provide better support for beginners in design and fabrication. We recognize that haptic feedback is an essential part of paper interfaces, and we are excited to discuss and envision this approach at SIGGRAPH 2024.

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REFERENCES

- Michelle Annett, Tovi Grossman, Daniel Wigdor, and George Fitzmaurice. 2015. MoveableMaker: Facilitating the Design, Generation, and Assembly of Moveable Papercraft. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (UIST '15)*. Association for Computing Machinery, New York, NY, USA, 565–574. <https://doi.org/10.1145/2807442.2807483>
- Sandra Bae, Ruhan Yang, Peter Gyory, Julia Uhr, Danielle Albers Szafir, and Ellen Yi-Luen Do. 2021. Touching Information with DIY Paper Charts & AR Markers. In *Interaction Design and Children (IDC '21)*. Association for Computing Machinery, New York, NY, USA, 433–438. <https://doi.org/10.1145/3459990.3465191>
- S. Sandra Bae, Rishi Vanukuru, Ruhan Yang, Peter Gyory, Ran Zhou, Ellen Yi-Luen Do, and Danielle Albers Szafir. 2022. Cultivating Visualization Literacy for Children Through Curiosity and Play. <https://doi.org/10.48550/arXiv.2208.05015>
- Zekun Chang, Tung D. Ta, Koya Narumi, Heeju Kim, Fuminori Okuya, Dongchi Li, Kunihiro Kato, Jie Qi, Yoshinobu Miyamoto, Kazuya Saito, and Yoshihiro Kawahara. 2020. Kirigami Haptic Swatches: Design Methods for Cut-and-Fold Haptic Feedback Mechanisms. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376655>
- Yoshihiro Kawahara, Steve Hodges, Benjamin S. Cook, Cheng Zhang, and Gregory D. Abowd. 2013. Instant inkjet circuits: lab-based inkjet printing to support rapid prototyping of UbiComp devices. In *Proceedings of the 2013 ACM international joint conference on Pervasive and ubiquitous computing (UbiComp '13)*. Association for Computing Machinery, New York, NY, USA, 363–372. <https://doi.org/10.1145/2493432.2493486>
- Jie Qi and Leah Buechley. 2010. Electronic popables: exploring paper-based computing through an interactive pop-up book. In *Proceedings of the fourth international conference on Tangible, embedded, and embodied interaction (TEI '10)*. Association for Computing Machinery, New York, NY, USA, 121–128. <https://doi.org/10.1145/1709886.1709909>
- Ruhan Yang. 2024. Paper Modular Robot: Circuit, Sensation Feedback, and 3D Geometry. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '24)*. Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3623509.3634899>
- Clement Zheng, HyunJoo Oh, Laura Devendorf, and Ellen Yi-Luen Do. 2019. Sensing Kirigami. In *Proceedings of the 2019 on Designing Interactive Systems Conference (DIS '19)*. Association for Computing Machinery, New York, NY, USA, 921–934. <https://doi.org/10.1145/3322276.3323689>